

KUSH PATEL

732-421-8945 | kushpate@andrew.cmu.edu | [linkedin.com/in/patelkush1508](https://www.linkedin.com/in/patelkush1508) | github.com/kushp15 | US Citizen

Education

Carnegie Mellon University - GPA: 3.73/4.0

Pittsburgh, PA

Master of Science in Artificial Intelligence Engineering

May 2027

CourseWork: AI Systems & Toolchains, Machine Learning, Advanced NLP (PhD), Deep Learning, ML with Large Datasets, Trustworthy AI, Distributed Systems

Rutgers University - New Brunswick - GPA: 3.67/4.0

Bachelor of Science in Computer Science & Minor in Business Administration

Sep 2022 – Aug 2024

CourseWork: Computer Algorithms, Artificial Intelligence, Deep Learning, Operating Systems, Systems Programming

Technical Skills

Languages: Python, Go, C/C++, TypeScript, SQL

AI/ML: LLMs, RAG, AI Agents, MCP, Transformers, QLoRA, Quantization, OCR, Computer Vision

Technologies: Postgres, Kafka, BigQuery, dbt, FAISS, Docker, Git, CI/CD, AWS, GCP, Azure

Frameworks: PyTorch, TensorFlow/Keras, Hugging Face, LangChain, LangGraph, JAX, ONNX, OpenCV, YOLO, Spark, Polars, FastAPI, Flask, React

Experience

Advantest America, Inc

South Plainfield, NJ

Software Engineer

Jun 2024 – Present

- Built an AI agent that reads emails and attached docs, extracts key requirements using an LLM-based parsing pipeline, and auto-updates version-controlled project files, **achieving 94% extraction accuracy across 1000+ documents**
- Developed an OCR inspection tool to track PCB layer stack-up, self-calibrate to each job, and alert operators to sequencing errors, **catching ~9 errors monthly and saving an estimated \$1M annually in scrap and rework**
- Designed ETL pipelines using Apache Spark and dbt with CI/CD-automated testing to clean and transform 500M+ records, **powering production reporting and analytics at 99% uptime**
- Connected robotic APIs to legacy measurement software by building a TCP/IP communication bridge, **automating 80% of the inspection workflow**
- Deployed a full-stack dashboard platform with FastAPI, caching, and optimized data processing, giving the team real-time visibility into live manufacturing production data

Technology Intern

Aug 2022 – May 2024

- Secured access for **500+ users** by configuring Azure AD IAM, MFA, and conditional access policies
- Automated audits and backup verification across **1,000+ endpoints**, streamlining compliance reporting

Technical Projects

Prompt Injection Defense for LLM Agents | CMU | LLM Security, AI Agents, RAG | [Report](#)

2026

- Built a 3-layer defense pipeline for LLM agents using structural delimiters, a router LLM firewall, and an LLM-as-a-judge, **cutting prompt injection success on InjecAgent from 75.0% to 4.5%**
- Added a deterministic symbolic verifier and TF-IDF RAG threat intelligence to catch attacks the LLM judge missed, **intercepting 97.7% of residual failures**

Efficient LLM Reasoning with Early Exit | CMU | Chain-of-Thought, LLM Inference | [Report](#)

2026

- Trained a lightweight MLP probe on 4,096-dimensional LLM hidden states to predict when the model is ready to answer, enabling early exit from chain-of-thought **with ~0.80 ROC-AUC**
- Built distributed inference and evaluation pipelines for long reasoning workloads, **achieving up to 98.1% simulated token savings across four benchmarks**

RAG Question-Answering System | CMU | Retrieval, Re-ranking, Web Crawling | [GitHub](#)

2026

- Developed a data pipeline to crawl, clean, and chunk web pages on CMU and Pittsburgh culture into a searchable knowledge base for question answering
- Built a hybrid retrieval pipeline combining BM25 and MiniLM dense embeddings with RRF and cross-encoder re-ranking, evaluated with F1, ROUGE, and LLM-as-a-judge, **improving F1 by 10% over BM25**